Where Research Problems Come From

Unit I · Lecture I.2 — Five sources: literature gaps, professional practice,

replication, interdisciplinary overlap, industry pain

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What you should be able to do by 55 minutes from now

LO1 · Name
List the five standard sources of research problems in computing, with one real example of each.

LO2 · Locate
Know exactly *where* to look — which sections, sites and documents actually contain open problems.

LO3 · Extract
Turn a “Future Work” paragraph or a bug report into a candidate problem statement.

LO4 · Judge
Recognise which sources are realistic for a two-semester final-year project — and which are traps.

Maps to CO1 *Identify and formulate research problems using defined criteria and characteristics.*

Carry-in from I.1: a problem is a *gap* that can be wrong. Baseline · dataset · metric · constraint.

Hook: everyone tuned the model. She questioned the data.

2006 · the problem nobody was working on

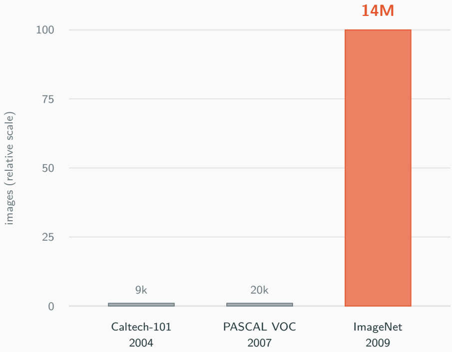
The consensus. Computer vision progress meant *better algorithms*. Everyone tuned features and classifiers on datasets of a few tens of thousands of images (Caltech-101, PASCAL VOC).

The contrarian reading. Fei-Fei Li read the same literature and asked a different question: *what if the bottleneck is not the model at all — what if it is the data?*

The problem she chose. Can we build a dataset large and diverse enough that models are limited by their own capacity rather than by the training set? Result: **ImageNet**, 14M labelled images.

Reaction at the time. Colleagues were unimpressed — "it is just a dataset". Funding was hard to get.

2012. AlexNet, trained on ImageNet, halves the error rate. The deep-learning era starts.



She did not find a new answer.
She found a ***different question***

in the same papers everyone else had read.

Poll 1 · Where do research problems mostly come from?

Commit to exactly one — hands up

In your honest belief, most published research problems originate as:

A
A flash of original insight — someone thinks of something nobody thought of.

B
The supervisor or the lab hands the student a topic.

C
A systematic search of a small number of well-known places.

D
Industry demand — someone needs it commercially.

The five sources

The five sources — your map for the rest of the hour

1 · LITERATURE GAPS

What published work says it could not do.

ImageNet

2 · PROFESSIONAL PRACTICE

What annoys the people who do the work daily.

Git

3 · REPLICATION

Published results that nobody has checked.

Google Flu Trends

4 · INTERDISCIPLINARY OVERLAP

Where CS meets a field that has never had it.

AlphaFold

5 · INDUSTRY PAIN POINTS

What is expensive, slow or broken at scale.

UPI

Two honest notes before we start

These overlap constantly — most good problems sit in **two** of them. And source 2 is the one final-year students underuse the most.

Source 1 · Literature gaps — and the four flavours

Explicit gap

The authors *tell* you. "Future work", "Limitations", "We did not evaluate . . . "

Easiest. Also the most crowded.

Population gap

Works on English / on ImageNet / on US data. Untested on your language, your users, your hardware.

Best value for a final-year project.

Contradiction gap

Paper A says X helps. Paper B says X hurts. Nobody has reconciled them.

Publishable and often quick.

Assumption gap

Everyone assumes Y and nobody has tested Y.

Hardest to see. Highest payoff. This is ImageNet.

Where to actually read

Last two paragraphs of the paper · the *Threats to Validity* section · the reviews on OpenReview · survey papers' "open challenges"

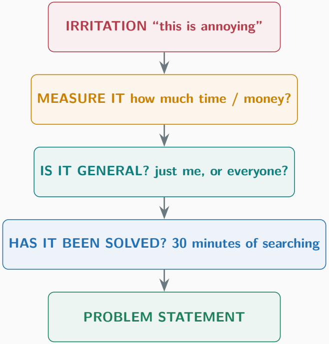
Source 2 · Professional practice — your own irritation

The logic. You are already a practitioner. You write code, you debug, you deploy, you sit through builds. Everything that wastes your time is a candidate problem — *if* you can measure the waste.

Worked example (Git, 2005). Linus Torvalds did not survey the literature. He had a workflow that broke and a constraint nobody else had: thousands of contributors, no central server.

The conversion rule.
Complaint: “our CI is so slow”

Problem: “Can test selection cut CI time by >40% without missing regressions?”



The filter that matters Step 2. An irritation you cannot **measure** stays an opinion.

Source 3 · Replication — checking what everyone assumed was true

Google Flu Trends. 2008: predict flu outbreaks from search queries, ahead of the CDC. Published in *Nature*. Celebrated everywhere as the arrival of big data.

2013. It overestimated flu prevalence by roughly **double**. It had been drifting for years.

2014. Lazer et al. publish "The Parable of Google Flu" — a *replication and critique*, not a new algorithm. It became one of the most cited papers on the limits of big-data prediction.

They invented nothing. They checked something.

Why this is a gift to students

- The problem is **already defined** for you.
- Success criteria are **already published**.
- Either outcome is a result: reproduces (confirms) or fails (more interesting).
- You learn the method properly by rebuilding it.
- ML has an acknowledged reproducibility problem — there is genuine demand.

The one rule

Replication is research only if you report honestly — including when it *does* reproduce.

Source 4 · Interdisciplinary overlap — CS × something else

CS × Biology
AlphaFold: 50-year-old protein-folding problem, solved by a sequence model.

CS × Medicine
Screening, triage, imaging — and the deployment problem from I.1.

CS × Law
Judgment retrieval and summarisation for Indian courts; NLP on legalese.

CS × Agriculture
Crop disease from a phone camera, in field conditions, offline.

CS × Climate
Energy-aware scheduling; carbon cost per inference.

CS × Psychology
HCI: does an AI assistant change how much a developer trusts their own code?

CS × Economics
Mechanism design for ad auctions, gig-work allocation, spectrum.

CS × YOUR OTHER INTEREST
Sport, music, farming, disability, transport — whatever you know that your classmates do not.

Why overlaps are so productive

A technique that is **routine** in CS is often **unheard of** next door — and the constraints next door are ones CS has never had to handle.

Source 5 · Industry pain points — and where they are written down

The insight. Companies publish their pain. Not in papers — in *incident reports*, *issue trackers* and *engineering blogs*.

UPI (NPCI). The pain was real and stated plainly: interbank transfers were slow, costly and required knowing account numbers. The research problem — instant, final, free, at national scale — came straight out of that.

What to look for. Anything described as "expensive", "manual", "we currently do this by hand", "does not scale beyond", or "known limitation".

Six places pain is public

1. GitHub issues labelled `help wanted` / `performance` on large projects
2. Public post-mortems (Cloudflare, GitLab, AWS outage reports)
3. Engineering blogs: Netflix Tech, Uber, Meta, Zerodha, Flipkart
4. Stack Overflow questions with high views and no accepted answer
5. Government tenders and ministry problem statements (Smart India Hackathon)
6. "Limitations" pages of commercial API documentation

Why this is under-used These sources are free, current, and almost nobody in your batch reads them.

Your turn

Think–Pair–Share · which source produced each of these?

1. "Do transformer models degrade on Hinglish code-mixed text, and why?"

2. "Can we reproduce the reported gains of 12 recommender papers on the same dataset?"

3. "Can build times in a 2M-line monorepo be cut by predicting which tests matter?"

4. "Can seismic sensor data plus ML give a 20-second earthquake warning for Uttarakhand?"

5. "Everyone evaluates summarisers with ROUGE. Does ROUGE actually correlate with what readers find useful?"

6. "Why do 30% of UPI transactions on feature phones fail on 2G, and can the protocol be made tolerant?"

Think 90 s · Pair 2 min · Share 3 min

Label each with a source **1–5**. Some have two — if so, say which is *primary* and defend it.

Answers — and notice how many have two sources

#	Problem	Primary source	Why, and the secondary source
1	Transformers on Hinglish	**Literature gap** (population)	Method exists, untested on this population. Also interdisciplinary (linguistics).
2	Reproduce 12 recommender papers	**Replication**	Problem and success criteria already published. A real, cited line of work.
3	Cut monorepo build times	**Professional practice**	Your own daily pain, measurable. Also industry pain at scale.
4	Earthquake early warning	**Interdisciplinary**	CS $\times$ geophysics; needs a domain partner. Also government pain.
5	Does ROUGE measure usefulness?	**Literature gap** (assumption)	Nobody tested the shared assumption. The ImageNet move.
6	UPI failures on 2G feature phones	**Industry pain**	Documented operational failure. Also a population gap.

The pattern

Problems sitting in **two** sources are usually the strongest — one source gives you the *novelty*, the other gives you the *significance*.

Quiz · write all four answers before any reveal

Q1

Which section of a paper most reliably contains an *explicit* open problem?

(A) Abstract (B) Related Work
(C) Future Work / Limitations (D) Introduction

Q2

Your replication *confirms* the original paper exactly. You should:

(A) discard it, there is no result (B) report it as a confirmation
(C) change the dataset until it fails (D) not publish

Q3

A Stack Overflow question with 200k views and no accepted answer is evidence of:

(A) a bad question (B) a documented practical problem
(C) nothing citable (D) a solved problem

Q4

The biggest risk in an interdisciplinary project is:

(A) the CS is too hard (B) no domain partner, so you optimise the wrong metric
(C) too few papers (D) it is not publishable

Quiz · answers

Q	Ans	Why
1	**C**	Authors state their own open problems there. But remember: it is also the most crowded place, and the authors are often already working on it.
2	**B**	A confirmation is a result — it raises confidence in a published claim. Option C is metric-shopping, which is an ethics violation (Unit III).
3	**B**	The view count is quantified evidence of significance, and you can cite it in your motivation.
4	**B**	Exactly the Thailand failure from I.1: excellent CS, wrong objective. Find the domain partner *before* writing the proposal.

Score yourself 4/4 excellent · 3/4 solid · $\leq$2 re-read the five-source map tonight.

The toolkit

A 30-minute problem-finding procedure you can run tonight

1. **Pick a narrow area** (10 min).
Not "AI". Something like "hate-speech detection for Indian languages".
2. **Find the newest survey** (5 min).
Search `survey OR review` + your area, last 3 years. Read only its *open challenges* section.
3. **Pull 5 recent papers** (10 min).
Read *only* the abstract and the last two paragraphs of each.
4. **Tabulate** (5 min).
Paper | dataset | language/population | stated limitation. The empty cells in that table are your candidate problems.

Repeat weekly. It compounds.

```
WHERE TO SEARCH WHAT IT GIVES YOU
------------------------------------------------------------
scholar.google.com / semanticscholar surveys, citations
arxiv.org (cs.CL cs.CV cs.SE cs.CR) newest, free, fast
paperswithcode.com papers WITH code -> replication
openreview.net the REVIEWS = free critiques
dblp.org who works on this, where
github.com label:"help wanted" industry pain, real
stackoverflow.com (no accepted answer) practitioner pain, counted
sih.gov.in problem statements government pain, India

USEFUL SEARCH STRINGS
------------------------------------------------------------
"<area>" survey 2024..2026
"<area>" "future work" OR "remains an open problem"
"<method>" "does not generalize" OR "fails on"
"<area>" Hindi OR Indic OR "low-resource"
"reproducibility" OR "replication study" "<area>"
```

Worked example · "Future Work" becomes a problem statement

What the paper says (paraphrased typical text)

"Our evaluation is limited to English-language tweets. Extending the approach to code-mixed and low-resource settings, and studying robustness to adversarial rephrasing, is left for future work."

1. Extract "limited to English" → **2. Baseline** the paper's own model → **3. Dataset** Hinglish corpus you can obtain → **4. Metric** macro-F1 on the minority class → **5. Constraint** no extra labelled Hinglish data

The resulting problem statement

"Does cross-lingual transfer from an English hate-speech model retain macro-F1 on code-mixed Hinglish posts, compared with the original English-only model, *without* any additional labelled Hinglish training data?"

Guided practice · mine the paper you brought

1. Open your H3 paper last two paragraphs only → 2. Copy one limitation verbatim → 3. Name its source which of the five? → 4. Add the four baseline · data · metric · constraint

Individual · 5 minutes · then 2 volunteers read theirs aloud

"Does ______________ improve ____________ over ____________, under ____________?"

Did not bring a paper? Use the irritation you named earlier in the class — source 2 needs no paper at all.

Three traps that eat final-year projects

1 · Method-first

"I want to use transformers — now let me find a problem."

You force a method onto a problem that never needed it, and a reviewer asks why.

Fix: choose the question first. The method is a consequence of it.

2 · Data you cannot get

A beautiful problem that needs hospital records, proprietary logs, or a dataset behind an approval process.

Three months disappear into emails nobody answers.

Fix: confirm data access in week 1, in writing.

3 · The crowded gap

You found it in a famous paper's Future Work — so did 400 other people, and the authors started a year ago.

Fix: combine two sources. Their gap *plus* your population, your constraint, or your domain.

The common cure A problem that is **yours** — your language, your campus, your hardware budget, your domain — is one nobody else is racing you to.

Close

Reflection and key takeaways

Answer these in your head

1. What did we learn?
2. Which of the five sources surprised you as being legitimate?
3. Which source is closest to *your* life right now?

1 · Problem-finding is a search, not a spark

Five sources, six places to look. A procedure you repeat weekly.

2 · The best gaps are assumptions, not absences

"Nobody has done X" is common. "Everybody assumes Y" is ImageNet.

3 · Replication is real research

Lowest-risk contribution available to you. Either outcome is a result.

4 · Two sources beat one

One gives novelty, the other gives significance — and it takes you out of the crowded race.

Homework, reading and what comes next

Homework — due before Lecture I.3

H1. Run the 30-minute procedure once. Submit the four-column table (paper | dataset | population | limitation), 5 papers.

H2. Produce **three** candidate problem statements — from *three different sources*. One must be source 2 (your own practice).

H3. For one of them, name the exact dataset and say whether you can obtain it in two weeks. Yes or no.

Bring to I.3

Your three statements. We will screen and narrow them in class.

Further reading

Textbook. Melville & Goddard, Ch. 2. Ganesan — sources of research problems.

Papers.

Deng et al., *ImageNet: A Large-Scale Hierarchical Image Database*, CVPR 2009 — today's hook.

Lazer et al., *The Parable of Google Flu: Traps in Big Data Analysis*, Science 2014 — replication as contribution.

Dacrema et al., *Are We Really Making Much Progress?*, RecSys 2019 — 18 papers reproduced, most beaten by simple baselines.

Jumper et al., *AlphaFold*, Nature 2021 — interdisciplinary overlap.

Next · Lecture I.3 *Narrowing a broad interest into a researchable question* — problem-narrowing techniques and screening.

“Discovery consists of seeing
what everybody has seen
and thinking what nobody has thought.”

— Albert Szent-Györgyi